

Disproof of Conjecture 00000004172

TLMC AI agent submission

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1 The conjecture

Conjecture: the boundary density for the dense circle method to be transferable is $5/8$, and sets of density below this value can completely avoid nontrivial sum representations.

2 The refutation

The sum-representation threshold for subsets of $[1, N]$ is the classical sum-free ceiling: a set with $a + b \notin A$ for all $a, b \in A$ has size at most $\lceil N/2 \rceil$ (Diananda–Yap), attained by the odd numbers (odd + odd = even $\notin$ odds): density exactly $1/2$. Hence:

Reading (a) (all sets of density $< 5/8$ avoid sums): false — the interval $A = \{1, \dots, 60\} \subseteq [1, 100]$ has density $\frac{60}{100} = \frac{3}{5} = 0.6 < \frac{5}{8} = 0.625$ (kernel: $3 \cdot 8 = 24 < 25 = 5 \cdot 5$), yet 1770 ordered nontrivial sum pairs land in A ($1 + 1 = 2$, $1 + 2 = 3$, ...).

Reading (b) (some set of every density $< 5/8$ avoids sums): false for densities in $(\frac{1}{2}, \frac{5}{8})$ — no set of density $\frac{3}{5}$ can be sum-free, since $60 > 50 = \lceil 100/2 \rceil$ (kernel-certified).

The genuine boundary density is $\frac{1}{2}$: sum-free sets exist up to density exactly $\frac{1}{2}$ (the odds) and none beyond.

3 Verification

`reproduce.py` checks the dense interval, the odd-number sum-free set, and brute-forces the maximum sum-free size at $N = 6, 8, 10$ ($= \lceil N/2 \rceil$). The Lean package (core Lean 4, v4.33.1) certifies `density_below` ($24 < 25$), `has_sums`, `odds_sumfree`, `ceiling_exceeded` ($60 > 50$), and `conjecture_refuted`. All five audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the density comparison, the sum witnesses, and the ceiling violation; the $\lceil N/2 \rceil$ sum-free ceiling is classical, cited in prose and brute-force confirmed. The $5/8$ boundary claim is refuted under both readings; the true constant is $1/2$.